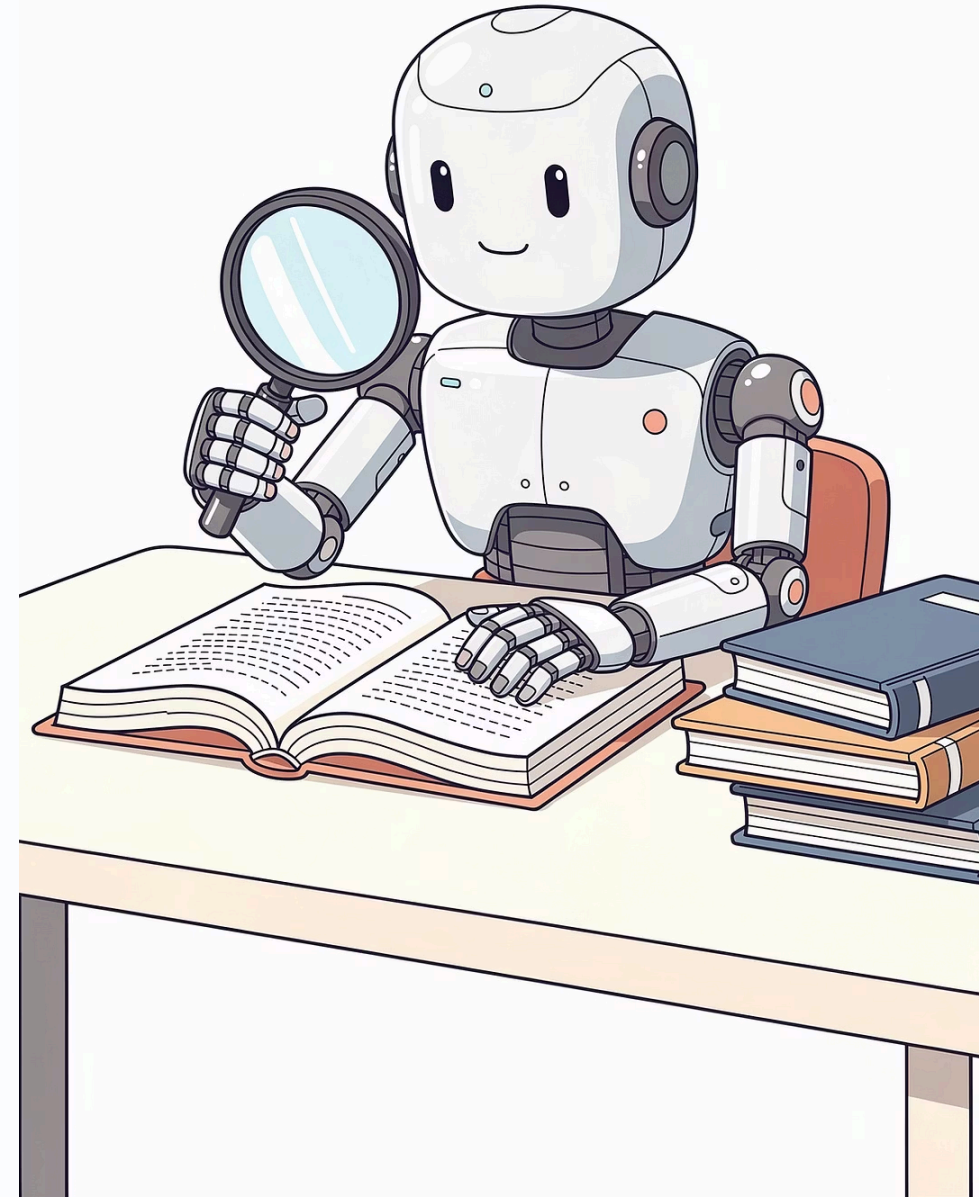


Session 14:

AI That Reads - Natural Language Processing (NLP)

Over the next 4 sessions, we'll look at how AI uses its different "senses" - reading, seeing, hearing, and combining them.

This Session: how does AI actually understand **words**? 📖



Objectives

By the end of this Session, you'll be able to:

01

Explain NLP

Describe what NLP is - and why it's so hard for computers!

02

Know the 3-Step Recipe

Master the **Break** → **Understand** → **Generate** framework AI uses to read.

03

Spot NLP Jobs

Recognize translation, summarizing, sentiment analysis, and more.

04

Try Real Tools

Use QuillBot, Google Translate, and NotebookLM hands-on.

05

Understand the Tricky Parts

Learn why language is genuinely hard for AI to understand.



Recap & Homework

Welcome back! Let's check in on your **AI Helper Week**. 📖

Reflection Questions

- How did your AI Helper Week go?
- Which day did AI help the most?
- Did you add anything to your Prompt Library?

Quick Check from Last Session

- What's the right tool for **research with sources**?
- What's the right tool to **polish YOUR writing**?

Warm-Up - Language Is Tricky! 🤔

Look at these sentences. Are they easy or **HARD** to understand?

"I saw a man on a hill with a telescope." 📡

Who has the telescope - the man, or me? The sentence doesn't say!

"Time flies like an arrow."

Is "flies" a verb? Or are there bugs called "time flies"? Both readings are grammatically valid!

"Oh great, another pop quiz."

Is this person actually happy? Or being sarcastic? You can tell - but can a computer?

- ① You get it instantly. But how would a computer figure it out? That's exactly what NLP tries to solve.

What is NLP?

NLP (Natural Language Processing) is a fancy term for: **AI that works with human words.**

Reading

AI takes in written or spoken human language.

Understanding

AI figures out what those words actually mean.

Writing Back

AI generates a useful response in human language.

Everywhere you see AI working with text - that's NLP.

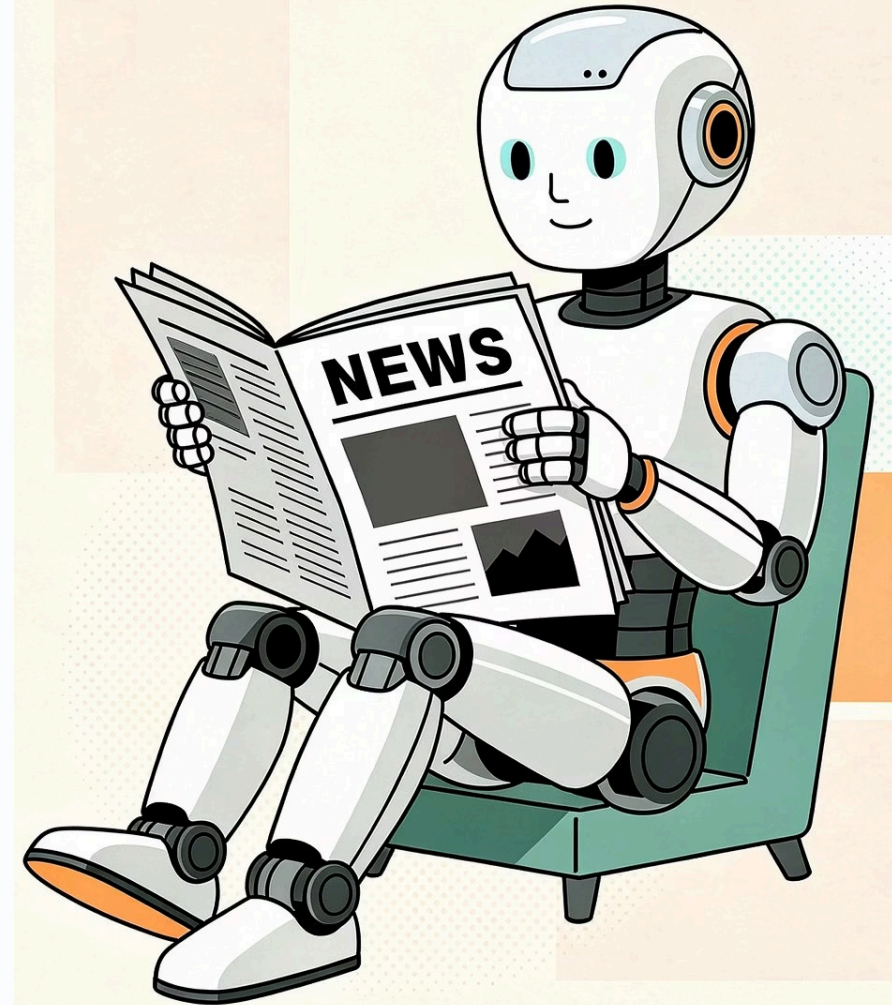
ChatGPT

Siri

Google Translate

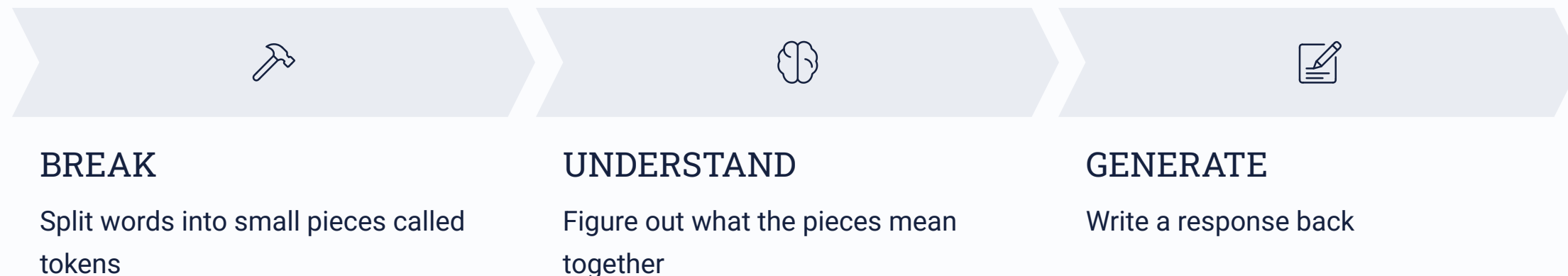
YouTube auto-captions

Autocomplete on your phone



The 3-Step NLP Recipe

Most NLP follows 3 simple steps - and everything AI does with language comes back to this:



✓ **Break → Understand → Generate.** That's it. That's how AI reads.

Every NLP tool you'll ever use is built on this recipe.

Step 1 - BREAK 🛠️

AI can't read whole sentences at once. It breaks them into small pieces called **tokens**.

Example: "I love pizza!"

AI breaks it down into 4 tokens:

"I"

"love"

"pizza"

"!"

Think of it like LEGO - break the sentence into bricks first, then you can work with each piece.

AI Also Labels the Pieces

After breaking, AI labels what each piece **IS**. Knowing the type of each word helps AI figure out what the sentence **MEANS**.

Example: "The cat sleeps."



"The" → Descriptor

Describes a noun - tells us which one



"cat" → Noun

A thing - the subject of the sentence



"sleeps" → Verb

An action - what the cat is doing

❗ Without labels, AI would just see a jumble of words.

With labels, it knows what's an action, what's a thing, and what's a description.

AI Spots Names and Places Too

AI can spot **special words** like names, places, and dates.

This is called **Named Entity Recognition (NER)**.

Example: "Sara visited Tokyo last summer."

Sara

Recognized as a
person

Tokyo

Recognized as a
place

Last summer

Recognized as a
time

Fancy name, simple idea: AI knows which words are **people, places, or times**. This is how Google can show you maps when you mention a city.



Step 2 - UNDERSTAND

Now AI has the pieces. Time to figure out what you **mean**. This step has an official name: **NLU (Natural Language Understanding)**.



Intent

What do you **want**? Are you asking a question? Giving a command? Making a joke?



Sentiment

What's the **feeling**? Happy? Sad? Angry? Excited? Sarcastic?



Connections

How do words **relate**? Who did what to whom? What modifies what?

① AI sounds like it understands you, but it's really pattern-matching what humans usually want.

Understand in Action - Two Examples

Let's see the UNDERSTAND step working on two real examples:

Example 1: A Voice Command

"Set a reminder to call grandma tomorrow at 5pm."

Intent: set a reminder

Action: call grandma

Time: tomorrow at 5pm

Who: grandma

Example 2: Detecting Feelings

"Ugh, my team lost AGAIN today."

Sentiment: negative (sad/frustrated)

Topic: your team

When: today

👉 Same step, different jobs. **Understanding = figuring out the meaning behind the words.**

Mini Activity 1 - Break a Sentence!

Let's do what AI does. Break this sentence into tokens:

"My dog Bella loves to chase squirrels in the park."

Spot the Special Words (NER)

 **Person/Pet**
Who is the named character? _____

 **Place**
Where does the action happen? _____

 **Action**
What is the main verb? _____

Label These Tokens

Token	Label (Noun / Verb / Descriptor)
"My"	?
"loves"	?
"squirrels"	?

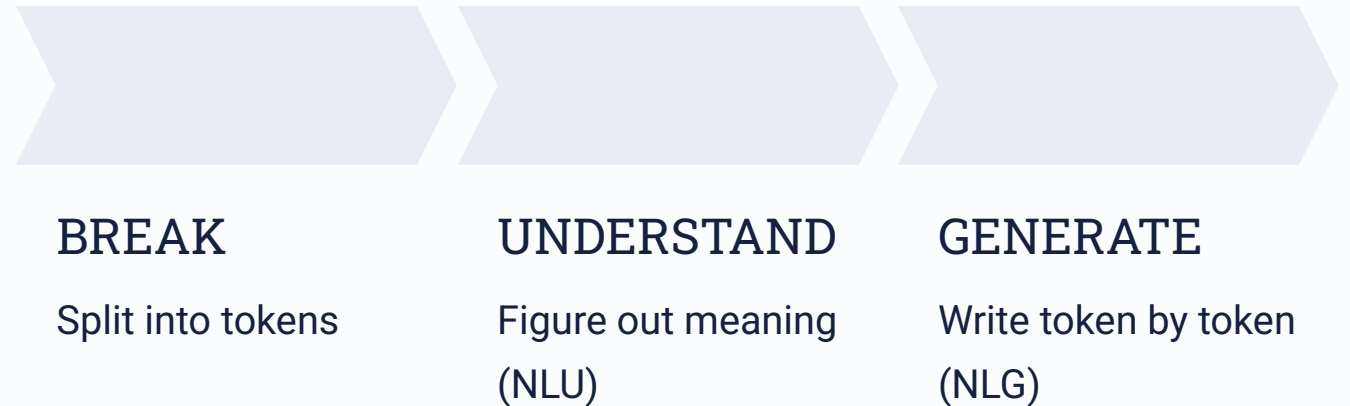
 Hint: think NOUN, VERB, or DESCRIPTOR



Step 3 - GENERATE 🖋️

Finally, AI writes a response back (if needed).

You say: "What's a fun fact about giraffes?" AI generates: "Giraffes... have... the... longest... neck... of... any... mammal!"



🧱 Token by token - building the sentence like LEGO.

This step is **NLG (Natural Language Generation)**.

Common NLP Jobs

NLP does **tons** of useful jobs. Here are the big ones - all built on Break → Understand → Generate:

Job	What it does	Example
 Translate	One language to another	"Hello" → "Hola"
 Summarize	Make long stuff short	10-page article → 3 sentences
 Paraphrase	Rewrite the same thing differently	"I'm tired" → "I feel exhausted"
 Sentiment	Is this happy or sad?	"I LOVE this!" → 😊 happy
 Answer Questions	Find answers to your questions	"When did WWII end?" → "1945"
 Find Names/Places	Spot special words	"John lives in NYC" → 👤 John, 🏙️ NYC

👉 The first 3 - Translate, Summarize, and Paraphrase - are what we'll try hands-on today!

Mini Activity 2 - Match the NLP Job!

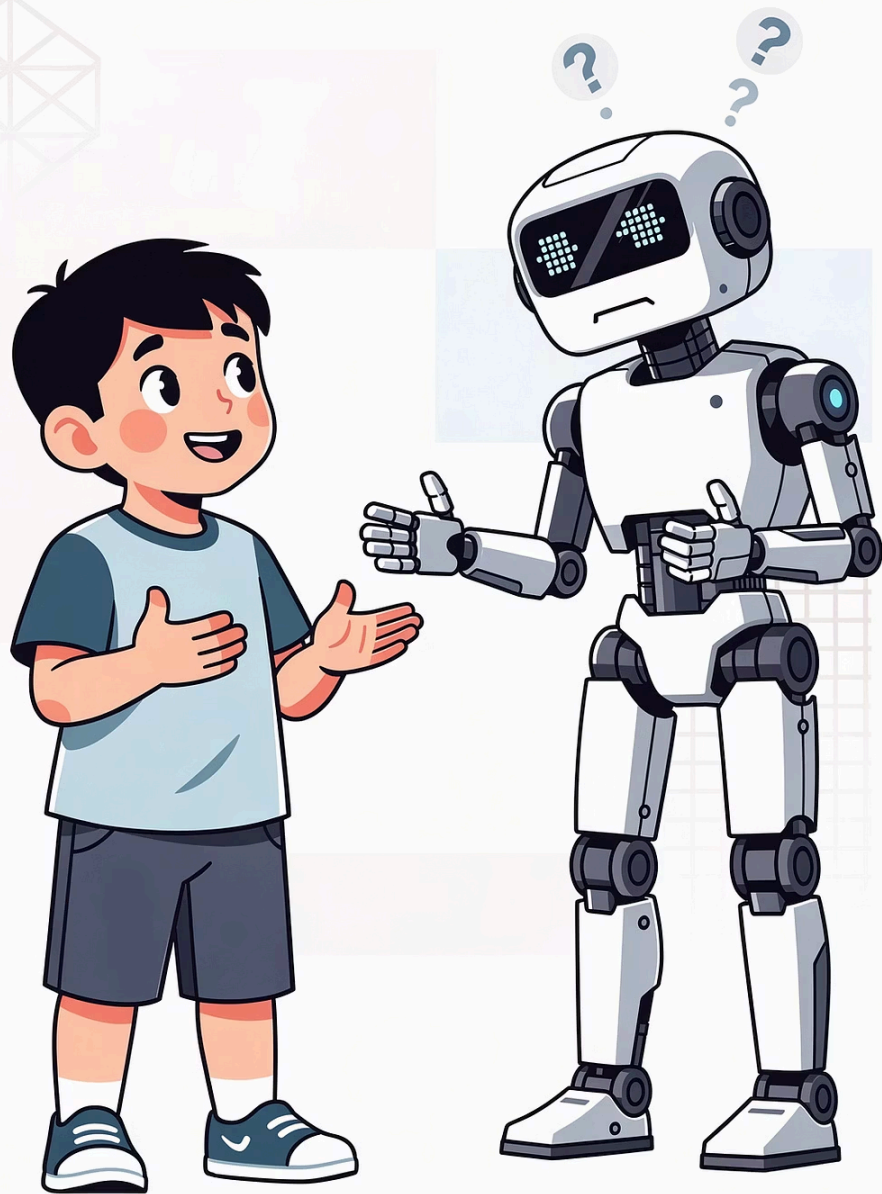
Look at each example. What NLP job is happening? Try to match before looking at the answers!

What's happening	NLP Job?
Translating a French menu to English on your phone	?
YouTube auto-captions when you watch a video	?
Gmail flagging an email as spam	?
Google summarizing a long article at the top of search	?
Snapchat detecting "happy" in your text reply	?
Wikipedia detecting "Paris" as a place	?

 **Options:** Translate / Speech-to-Text / Spam Detection / Summarize / Sentiment / Named Entity Recognition

Mini Activity 2 - Answers

What's happening	NLP Job
Translate French menu	 Translate
YouTube auto-captions	 Speech-to-Text (<i>uses Break → Understand → Generate on SOUND instead of text!</i>)
Gmail spam detection	 Sentiment / Classifying
Google article summary	 Summarize
Snapchat detecting "happy"	 Sentiment
Wikipedia detecting "Paris"	 Named Entity Recognition



Why is Language HARD for AI?

Even with the 3-step recipe, AI struggles with these four challenges - all happening in the **UNDERSTAND** step:



Ambiguity

"I saw her duck." - Did she dodge? Or do you mean her pet duck?



Sarcasm

"Oh wonderful, more homework." - Actually NOT wonderful!



Idioms

"It's raining cats and dogs!" - No actual cats. No actual dogs.



Different Cultures

"Football" in the US ≠ "Football" in most of the world ⚽

AI can **BREAK** any sentence into tokens. AI can **GENERATE** responses. But **UNDERSTANDING** what humans actually mean? That's where AI struggles.

Hands-On Activity - Try 3 NLP Tools!

Time to actually **USE** NLP tools. Today we'll try 3 real tools - one for each of 3 different NLP jobs.

 Google Translate

NLP Job: **Translation**

 NotebookLM

NLP Job: **Summarizing**

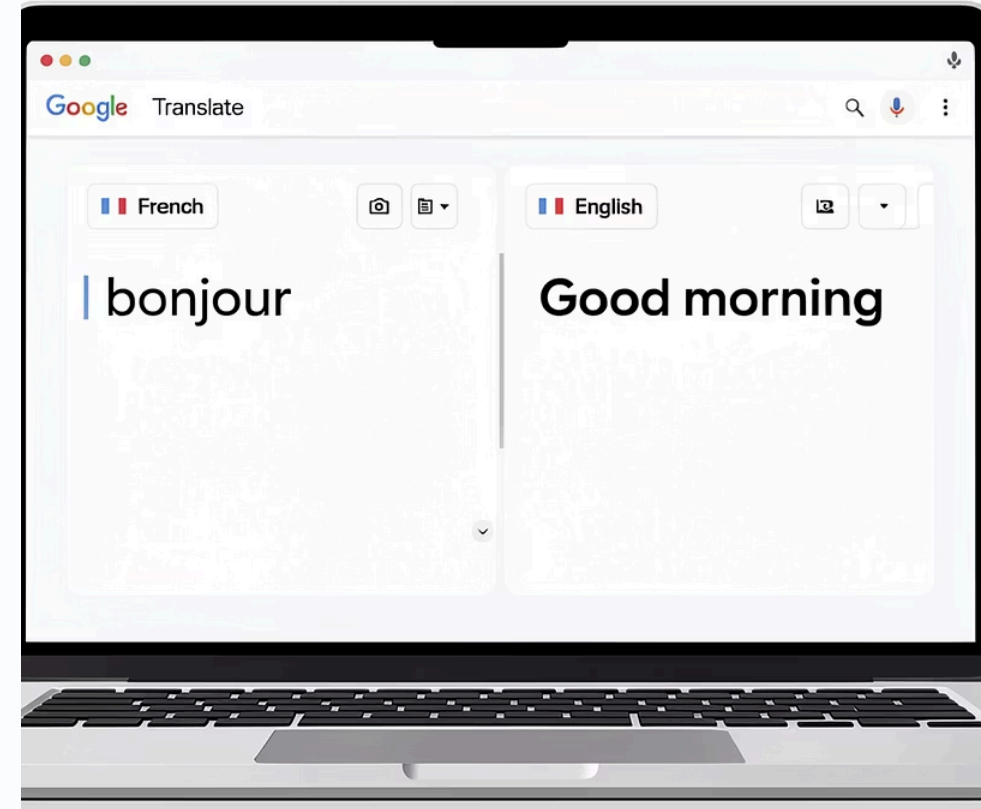
 QuillBot

NLP Job: **Paraphrasing**

① Three different tools. Three different NLP jobs. **All built on the same Break → Understand → Generate recipe.** Watch for the recipe in each tool!

Google Translate

- 1 Open translate.google.com
Head to translate.google.com in your browser.
- 2 Type the phrase
Enter: *"Hello, how are you today?"*
- 3 Translate to Spanish, then Japanese
What do you get each time? Are they different?
- 4 Tricky test: try an idiom!
Type *"It's raining cats and dogs!"*
Did AI understand it, or translate word-by-word?





NotebookLM

Summarizing uses Understanding to get the main ideas, then Generating to write the shorter version.

1 Open notebooklm.google.com

Head to notebooklm.google.com in your browser.

2 Add a source

Could be a short Wikipedia article - like about your favorite animal!

3 Ask for a summary

Click "Summarize" or ask: *"Summarize this in 3 sentences for a 5th grader."*

4 Read and reflect

Did NotebookLM **shorten** the article but keep the main ideas? That's summarizing!

QuillBot

Paraphrasing = Understanding what you said + Generating a new way to say it.

1 Open quillbot.com

Head to quillbot.com in your browser.

2 Write your own sentence

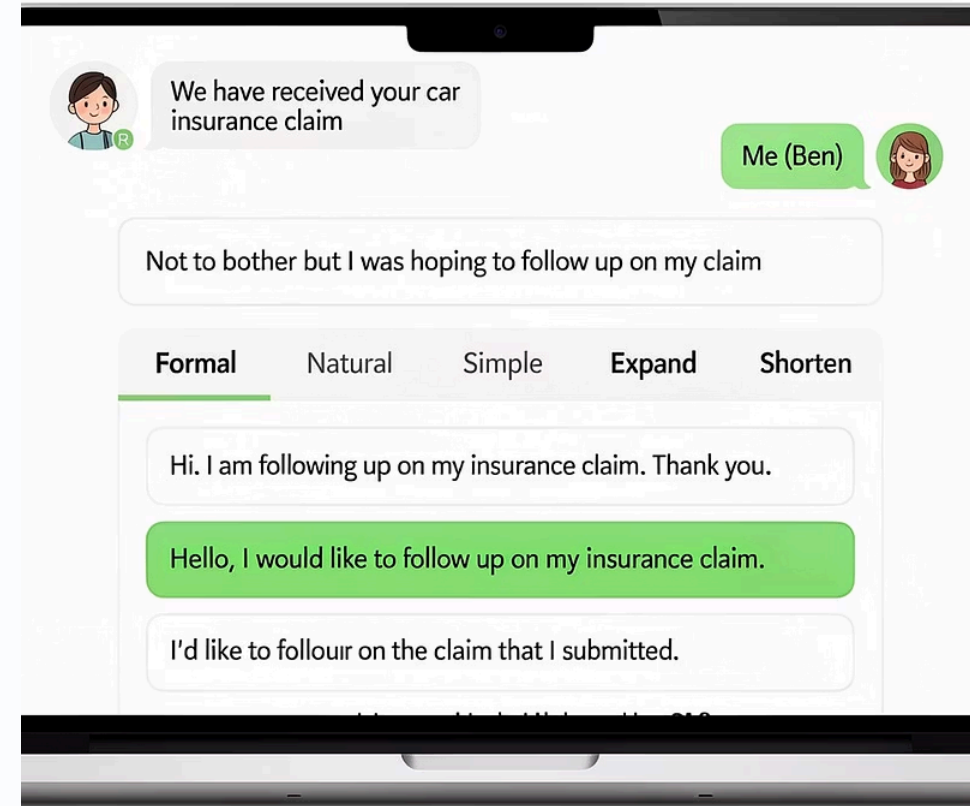
Something simple like: *"I really like my dog because he is fun and cute."*

3 Click "Paraphrase"

Did QuillBot keep the **meaning** but change the **words**? That's paraphrasing!

4 Try different modes

Switch between Standard, Fluency, and Creative. Which sounds best to you?





What Did You Notice?

Discussion time! Reflect on your experience with all three tools.

 **Most Useful?**

Which tool was the most useful **for YOU**? Why?

 **Surprised?**

Did any of them mess up or surprise you? What happened?

 **Could You Trick Them?**

Did you find a way to confuse any of the tools?

What We Learned in this Session

You now know how AI reads. 📖 Here's everything we covered:

- NLP = AI working with human words
Reading, understanding, and writing back in human language.
- Tokens = small word pieces AI uses
AI breaks every sentence into tokens before processing.
- Language is HARD for AI
Sarcasm, idioms, and ambiguity all confuse the Understand step.
- 3-Step Recipe: Break → Understand → Generate
Bonus vocab: Understand = **NLU**. Generate = **NLG**.
- Common NLP jobs
Translate, Summarize, Sentiment, Paraphrase, Q&A, Find Names.
- Tried 3 real NLP tools
Google Translate (Translate), NotebookLM (Summarize), QuillBot (Paraphrase).

Your Homework - NLP Detective

Find NLP in your daily life this week. **Mission:** Spot **3 examples** of NLP being used somewhere around you.

Examples to Look For

Auto-translate on a YouTube comment or webpage	An app summarizing something for you	An app detecting sentiment (Snapchat reactions, etc.)
Spell-check or autocorrect	Voice typing on a phone	News article summaries

Write Down Your Findings

What	NLP Job	Tool/App
1.		
2.		
3.		

Challenge

Try to **trick** an NLP tool!

Sarcasm Test 🤨

Type a **sarcastic** message in Google Translate. Does it translate the sarcasm correctly, or does it miss the tone?

Slang Test 👍

Use **slang** in QuillBot. Does it understand what you mean, or does it get confused?

Named Entity Test 🔍

Ask ChatGPT to find a **place** in a weird or tricky sentence. Does it find the right one?



Where do these tools fail? **Discuss in next session!**

Next Time Session 15:

AI That SEES - Computer Vision

You just learned how AI **reads**. Next session: how does AI **SEE**?

Face Recognition

How does AI recognize faces in photos and videos?

Self-Driving Cars

How do self-driving cars "see" the road and make decisions?

Cats vs. Dogs

You'll connect back to Session 6 - remember cats vs. dogs?

Tricky Challenges

Why is Computer Vision tricky for things like skin color and lighting?

